

Jayadratha Gayen

contactjayag@gmail.com | [Linkedin](#) | [Github](#) | [Google Scholar](#) | [Homepage](#)

EDUCATION

International Institute of Information Technology, Hyderabad
M.Sc in Electronics and Communication Engineering by Research (CGPA: 8.29/10)

Hyderabad, India
July 2022 - July 2025

Indian Institute of Engineering Science and Technology, Shibpur
Bachelor of Technology in Electrical Engineering (CGPA: 8.02/10)

Howrah, India
July 2014 - June 2018

RESEARCH EXPERIENCE

Research Fellow

Machine Learning Lab - MLL (IITH)

Nov 2022 - July 2025

Hyderabad, India

- worked in graph neural network (GNNs), uncertainty management, 3D vision
- Conducted research on GNNs, high-stakes applications like legal judgment prediction, disease prediction; accepted at TMLR 2025, DAI AAAI 2025
- Explored continuous time dynamic graphs (CTDGs), reject option classification; accepted in ASONAM 2025, TGL KDD 2025
- Worked on 3D computer vision application for pose transfer and motion transfer

PUBLICATIONS

1. **DevoTG: Temporal Graph Neural Networks for Modeling C. elegans Developmental Connectomics**
arXiv 2026 [[ArXiv](#), [Code](#)]
Jayadratha Gayen, Bradly Alicea
2. **Reducing Misclassification Risk in Dynamic Graph Neural Networks through Abstention**
Accepted at ASONAM 2025 [[ArXiv](#), [Code](#)]
Jayadratha Gayen, Himanshu Pal, Naresh Manwani, Charu Sharma
3. **Confidence First: Reliability-Driven Temporal Graph Neural Networks**
Accepted at KDD 2025 Temporal Graph Learning (TGL) Workshop [[Openreview](#), [ArXiv](#), [Code](#)]
Jayadratha Gayen, Himanshu Pal, Naresh Manwani, Charu Sharma
4. **Node Classification With Reject Option**
Accepted at TMLR 2025 [[Openreview](#), [ArXiv](#)]
Uday Bhaskar, Jayadratha Gayen, Charu Sharma, Naresh Manwani
5. **Node Classification With Integrated Reject Option For Legal Judgement Prediction**
Accepted at AAAI 2025 Deployable AI (DAI) Workshop [[ArXiv](#), [Slides](#)]
Uday Bhaskar, Jayadratha Gayen, Charu Sharma, Naresh Manwani

WORK EXPERIENCE

AI Research Engineer

CloudAngles Inc. (Codebenders)

Feb 2026 - Present

Hyderabad, India

- Building Codebenders, an agentic coding assistant: developed repository indexing, semantic retrieval, and context-management pipelines that facilitate natural language driven code editing, debugging, and generation.
- Cut AI token usage by **76.3%** and API costs by **64.2%** through in-session tool result caching and smarter tool loading.
- Improved response speed by **22.4%** with better context and memory management so the agent resumes exactly where it left off after any interruption.

Open Source Developer

Google Summer of Code at INCF

May 2025 - Sep 2025

Remote

- Developed **DevoTG**, a temporal GNN framework for modeling C. elegans embryogenesis and connectome formation; work published on [arXiv \(2606.21940\)](#).

- Temporal approach achieved mean test AUC of **0.839 vs 0.577** for static methods, showing temporal memory as the decisive factor.
- Modeled spatiotemporal cell division and neuronal network formation as evolving graphs; built interactive 3D visualizations for biological data exploration.

Engineer

Vikram Solar Limited

July 2018 - Dec 2019

Falta, West Bengal, India

- Contributed to a Six Sigma project on rolled throughput improvement, increasing production by 10%.
- Did bottleneck analysis and DOE using Pareto, Five Why, and Fishbone methods to improve production quality.
- Analyzed technical data to decide current binning for all module types and introduced RFID tag automation, reducing manual errors. Utilized SAP and extensive Excel analysis for data management and insights.
- Led root cause analysis and KAIZEN projects, achieving a 2% reduction in power consumption.

SELECTED PROJECTS

Nyay Saheli: Conversational RAG for Indian Women [\[GitHub\]](#) 2025

- Developed a conversational AI assistant using Retrieval-Augmented Generation (RAG) to provide empathetic legal guidance to Indian women in multiple Indian languages.
- Implemented multilingual natural language processing capabilities to deliver culturally sensitive and legally accurate information for women's rights and safety.
- Designed system to handle complex legal queries, maintaining empathetic tone and cultural context for Indian communities.

Node Classification With Integrated Reject Option [\[Paper Link\]](#) 2024

- Introduced a novel graph neural network framework, NodeCwR, incorporating cost-based and coverage-based rejection strategies to enhance the reliability of Legal Judgment Prediction (LJP) and disease prediction tasks.
- Achieved 6-19% improvement in accuracy on the ILDC dataset for LJP task after abstaining from uncertain cases.
- Enhanced interpretability of the rejection decisions using SHAP visualizations.

Predict Confidently, Predict Right: Abstention in Dynamic Graph Learning [\[Paper Link\]](#) 2024

- Developed a novel reject option strategy for CTDGs, allowing the model to abstain from uncertain predictions, enhancing reliability in link prediction and node classification tasks
- Achieved 10-15% improvement in AUC/AP scores across six dynamic graph datasets, effectively addressing class imbalance and critical misclassification
- Demonstrated model effectiveness through experiments in risk-sensitive domains in fraud detection and disease prediction.

3D Motion Transfer Across Diverse Character Topologies [\[Project Output\]](#) 2023

- Developed a novel method using Graph Convolution Networks (GCN) for realistic 3D motion transfer between diverse character topologies, including humanoids and quadrupeds.
- By incorporating prior frame encodings, the approach reduces jitter and ensures smoother, natural animations.
- outperforms traditional pose transfer techniques, especially on complex characters, shown on the Mixamo dataset.

TEACHING EXPERIENCE

Teaching Assistant

3D Vision Summer School

May 2023

CVIT, IIIT Hyderabad, India

- Conducted a tutorial on graph neural networks, their current progress, and case studies on the usage of GNNs for 3D vision especially mesh-based systems.
- Taken doubt-solving sessions in GNNs and 3D vision concepts for the participants.

Instructor

Executive Training Program on AIML

October 2023

ihub-data, IIIT Hyderabad, India

- Coordinated in two-weeks executive training program on machine learning and artificial intelligence.
- conducted tutorials concepts like PCA, end-to-end ML model building. Also taken doubt-solving sessions.

TECHNICAL SKILLS

Languages: Python, C++, C, SQL

ML/DL: PyTorch, PyG, scikit-learn, TensorFlow, Keras, Transformers

AI/Agent: LangGraph, LangChain, RAG, Agentic AI Systems, LLM APIs, MCP

Tools: GitHub, VS Code, Conda, Poetry, Blender, MeshLab, Minitab, MATLAB Simulink

EXTRACURRICULAR & PARTICIPATION

- National Finalist (top 6/275+ teams), BharatGen Hackathon: Designed *Nyay Saheli*, a GenAI-powered multilingual chatbot offering personalized legal guidance for Indian women. (*Apr-June 2025*)
- **Founding Member, Srijani - The Hope:** Started an independent non-profit NGO focused on improving education and healthcare in remote areas of West Bengal, India, with limited accessibility. (*Mar 2020 - Apr 2022*)
- Participated in LoG Delhi Meetup, November 26, 2024
- Attended 8th Summer School on AI, IIIT Hyderabad, July 2024

REFERENCES

Below are individuals who can vouch for my skills, expertise, and contributions:

- **Prof. Charu Sharma (*Preferred*)** - Assistant Professor, Machine Learning Lab, IIIT Hyderabad - charu.sharma@iiit.ac.in
- **Prof. Anindita Sengupta** - Head, Electrical Engineering, IEST Shibpur - aninsen@ee.iests.ac.in
- **Dr. Bradly Alicea** - Head Scientist, Orthogonal Research and Education Laboratory; Lecturer, iSchool @ UIUC - balicea@openworm.org
- **Prof. Naresh Manwani** - Assistant Professor, Machine Learning Lab, IIIT Hyderabad - naresh.manwani@iiit.ac.in